Log-Hermite Functions, the Mellin Transform, and the Local Riemann Hypothesis

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August 2026

Abstract

The Mellin transform is the Fourier transform of the multiplicative group $\mathbb{R}_+$. Transporting the Hermite functions along the isomorphism $x = e^t$ yields the *log-Hermite functions*, an orthonormal basis of $L^2(\mathbb{R}_+)$ diagonalizing a canonical Mellin-type involution with eigenvalues $(-i)^n$. We record the dictionary, note that orthogonality survives the transport while polynomial completeness does not (the log-normal weight is moment-indeterminate; its orthogonal polynomials are Stieltjes–Wigert), and trace the connection to the Riemann zeta function: the functional equation of ξ is the Fourier-invariance of the Gaussian, and the Mellin transforms of the higher Hermite functions satisfy a provable "local Riemann Hypothesis" (Bump–Ng). We compute these Mellin transforms explicitly for $n \leq 6$ and exhibit their zeros on the critical line.

1 The dictionary: Mellin is Fourier on $\mathbb{R}_+$

Let $\psi_n(t) = c_n H_n(t)\, e^{-t^2/2}$, $c_n = (2^n n!\sqrt{\pi})^{-1/2}$, be the Hermite functions: an orthonormal basis of $L^2(\mathbb{R})$ satisfying $\widehat{\psi_n} = (-i)^n \psi_n$.

The substitution $x = e^t$ is an isomorphism $(\mathbb{R}_+, \times) \cong (\mathbb{R}, +)$ and induces a unitary

$$U : L^2(\mathbb{R}_+, dx) \to L^2(\mathbb{R}, dt), \qquad (Uf)(t) = e^{t/2} f(e^t), \tag{1}$$

the factor $e^{t/2}$ compensating the Jacobian. Under U the Mellin transform on the critical line becomes the Fourier transform:

$$\mathcal{M}f\big(\tfrac{1}{2} + i\tau\big) = \int_0^\infty f(x)\, x^{-1/2+i\tau}\, dx = \int_{\mathbb{R}} (Uf)(t)\, e^{i\tau t}\, dt. \tag{2}$$

The line $\operatorname{Re} s = \frac{1}{2}$ is thus the line of Mellin–Plancherel unitarity: it is Parseval's theorem read through U.

2 Log-Hermite functions

Define the *log-Hermite functions*

$$\varphi_n(x) = x^{-1/2}\, \psi_n(\log x), \qquad x \in \mathbb{R}_+, \tag{3}$$

so that $U\varphi_n = \psi_n$. Immediately:

- $\{\varphi_n\}_{n\geq 0}$ is an orthonormal basis of $L^2(\mathbb{R}_+, dx)$;
- by (2) and $\widehat{\psi_n} = (-i)^n \psi_n$,

$$\mathcal{M}\varphi_n\big(\tfrac{1}{2} + i\tau\big) = \sqrt{2\pi}\; i^n\, \psi_n(\tau) : \tag{4}$$

  the Mellin transform of a log-Hermite function is a plain Hermite function on the critical line;
- the harmonic oscillator $-\frac{d^2}{dt^2} + t^2$ transports to $-(x\frac{d}{dx})^2 + (\log x)^2 + \frac{1}{4}$ with eigenvalues $2n+1$; the Euler operator $x\frac{d}{dx}$ generates the dilations of $\mathbb{R}_+$ exactly as $\frac{d}{dt}$ generates the translations of $\mathbb{R}$.

Remark 2.1 (Eigenvectors, correctly stated). The equation "$\mathcal{M}\varphi_n = \lambda\varphi_n$" is ill-typed: $\mathcal{M}$ maps $L^2(\mathbb{R}_+)$ to L^2 of the critical line, a different space. To speak of eigenvectors one must identify the two spaces; the canonical choice $s = \frac{1}{2} + i\log x$ yields the unitary involution

$$(\mathcal{T}f)(x) = (2\pi)^{-1/2} \int_0^\infty (xy)^{-1/2}\, e^{\,i\log x \log y}\, f(y)\, dy, \tag{5}$$

which is $U^{-1}\mathcal{F}U$, satisfies $\mathcal{T}^4 = I$, and has

$$\mathcal{T}\varphi_n = (-i)^n \varphi_n. \tag{6}$$

The Fourier kernel e^{ixy} has become $e^{i\log x \log y}$, as it must. The distinction is invisible in the Fourier case only because $L^2(\mathbb{R})$ and its frequency space are silently identified.

Remark 2.2 (What does not transport). The weight e^{-t^2} becomes the log-normal density $e^{-(\log x)^2}/x$ on $\mathbb{R}_+$. This weight is moment-indeterminate (Stieltjes' original example). Its orthogonal polynomials are the Stieltjes–Wigert polynomials, a q-analogue of Hermite, and polynomials are *not* dense in L^2 of the log-normal weight. Orthogonality survives exponentiation; completeness of polynomials does not. The log-Hermite *functions* remain a complete basis regardless.

3 Connection to the Riemann zeta function

3.1 ζ lives on Mellin's unitarity line

With $\theta(x) = \sum_{n\in\mathbb{Z}} e^{-\pi n^2 x}$,

$$\pi^{-s/2}\,\Gamma\big(\tfrac{s}{2}\big)\,\zeta(s) = \int_0^\infty x^{s/2}\,\frac{\theta(x)-1}{2}\,\frac{dx}{x}. \tag{7}$$

The completed zeta function is the Mellin transform of a sum of Gaussians, and the critical line of the Riemann Hypothesis is the Mellin unitarity line of §1.

3.2 The functional equation is the Gaussian's eigenfunction property

Poisson summation applied to $e^{-\pi t^2}$ (which is fixed by $\mathcal{F}$) gives $\theta(1/x) = \sqrt{x}\,\theta(x)$, whence $\xi(s) = \xi(1-s)$. The chain is

$$\psi_0 \text{ Fourier-fixed} \implies \theta \text{ modular} \implies \xi(s) = \xi(1-s),$$

and $s \mapsto 1-s$ is reflection across the unitarity line.

3.3 The local Riemann Hypothesis

What do the higher ψ_n give? Bump and Ng showed that $\mathcal{M}\psi_n(s) = \int_0^\infty \psi_n(x)\, x^{s-1} dx$ factors as a zero-free Γ-factor times a polynomial $P_n(s)$ satisfying a functional equation under $s \mapsto 1-s$, and that *all zeros of* P_n *lie on* $\operatorname{Re} s = \frac{1}{2}$. This is the local Riemann Hypothesis (Bump–Ng; Bump–Choi–Kurlberg–Vaaler): provable for each Hermite function, open for the arithmetic sum $\theta - 1$.

4 Explicit computation for $n \le 6$

Applying $\int_0^\infty x^{k+s-1} e^{-x^2/2} dx = 2^{(s+k)/2-1}\,\Gamma\big(\frac{s+k}{2}\big)$ monomial by monomial to H_n and factoring out $\Gamma\big(\frac{s+\epsilon_n}{2}\big)$ ($\epsilon_n = n \bmod 2$):

$$\mathcal{M}\psi_n(s) = c_n\, 2^{(s+\epsilon_n)/2-1}\,\Gamma\Big(\frac{s+\epsilon_n}{2}\Big)\, P_n(s). \tag{8}$$

n	$P_n(s)$	P_n at $s = \frac{1}{2} + i\tau$	zeros (τ)
0	1	1	—
1	$2\sqrt{2}$	$2\sqrt{2}$	—
2	$2(2s-1)$	$4i\tau$	0
3	$4\sqrt{2}(2s-1)$	$8\sqrt{2}\,i\tau$	0
4	$4(4s^2-4s+3)$	$4(2-4\tau^2)$	$\pm 1/\sqrt{2}$
5	$8\sqrt{2}(4s^2-4s+7)$	$8\sqrt{2}(6-4\tau^2)$	$\pm\sqrt{3/2}$
6	$8(2s-1)(4s^2-4s+15)$	$16i\tau(14-4\tau^2)$	$0,\ \pm\sqrt{7/2}$

Since $4s^2 - 4s = 4(s-\frac{1}{2})^2 - 1 = -4\tau^2 - 1$ on the critical line, every $P_n(\frac{1}{2} + i\tau)$ is (up to a factor i^{ϵ_n}) a polynomial in τ with real coefficients, and its roots are real: all zeros lie on the line.

The symmetry (P_n real in τ) follows cheaply from $\widehat{\psi_n} = (-i)^n \psi_n$ and reality of ψ_n. The reality of the *roots* is the theorem: Bump–Ng prove it via a three-term recurrence, placing the P_n in orthogonal-polynomial territory where real interlacing zeros are the spectrum of a self-adjoint Jacobi matrix—a working miniature of the Hilbert–Pólya mechanism. Indeed the zeros spread and interlace with n: $\{0\} \subset \{\pm 0.707\}$, $\{\pm 1.225\} \subset \{0, \pm 1.871\}$.

5 Discussion

The pattern behind ζ: take f with clean Fourier behavior, sum over a lattice, apply Mellin. The eigenfunction property gives the functional equation for free; the open question is always whether the zeros stay on the unitarity line. For a single Hermite function they provably do. For $\theta - 1$, dilation $x \mapsto nx$ destroys the Hermite eigenstructure—Mellin turns dilations into the factors n^{-s} whose interference is the arithmetic difficulty.

Two open threads. First, since $\mathcal{M}$ is linear, the Mellin transform of any finite combination $\sum_m a_m \psi_m$ is a Γ-factor times a polynomial; which combinations preserve the local Riemann Hypothesis? The Hermite matching matrix (D_{mk}) of our companion paper is a distinguished change of basis between exactly such families. Second, the Stieltjes–Wigert polynomials suggest a q-deformation of the matching matrix, with cross- and self-pair weights becoming q-numbers.